

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – II

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Date of Assignment: 31-07-2026	Date/Time of Submission: 05-08-2026

Part A - Theory:

a) Connect to Different Data Sources (Excel, CSV, Database)

1. Open Tableau Desktop.
 2. Click **Connect**.
 3. Select **Microsoft Excel, Text File (CSV), or Database**.
 4. Choose the file or connect to the database.
 5. Click **Open**.
-

b) Create Data Extract vs Live Connection

1. Connect the data source.
 2. Select **Live** or **Extract**.
 3. Save the extract if required.
 4. Open a worksheet.
-

c) Manage Metadata

1. Open the **Data Source** tab.
 2. Right-click a field.
 3. Rename, Hide, or Change Data Type.
 4. Save the changes.
-

d) Perform Joins

1. Open the **Data Source** page.
2. Drag the second table.

3. Select the join field.
 4. Choose **Inner**, **Left**, or **Right** Join.
-

e) Apply Data Blending

1. Connect the first data source.
 2. Add the second data source.
 3. Select the common field.
 4. Tableau blends the data automatically.
-

f) Apply Filters

1. Drag a field to the **Filters** shelf.
 2. Select **Dimension**, **Measure**, or **Date** filter.
 3. Apply the required condition.
-

g) Optimize Performance

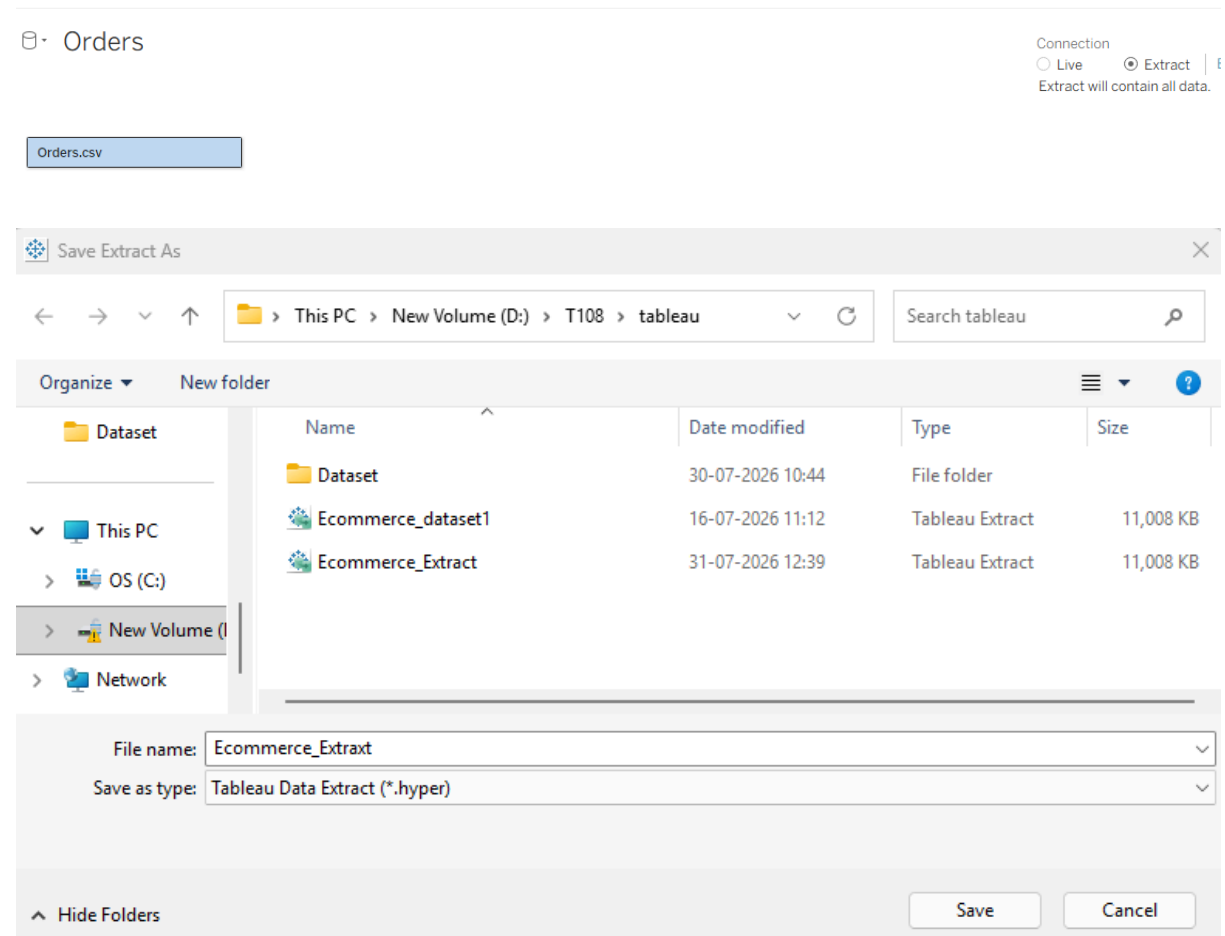
1. Use **Data Extract** instead of Live.
2. Hide unused fields.
3. Apply filters.
4. Save the workbook.

Part B – Program

Working with Data Sources and Data Preparation in Tableau

a) Connect to different data sources (Excel, CSV, Database)

b) Create and use data extracts vs live connections



c) Manage metadata (rename, hide, change data types)

Tables

Abc	Customer Id
	Order Approved At
	Order Delivered Carrier D...
	Order Delivered Customer...
	Order Estimated Delivery ...
Abc	Order Id
	Order Purchase Timestamp
Abc	Order Status
Abc	Measure Names

Tables

Abc	Customer Id
	Order Approved At
	Order Delivered Carrier ...
	Order Delivered Customer...
	Order Estimated Delivery ...
Abc	Order Id
	Order Purchase Timestamp
Abc	Status
Abc	Measure Names
#	Orders.csv (Count)
+	Measure Values

Tables

Abc	Customer Id
	Order Approved At
	Order Delivered Carrier D...
	Order Delivered Customer...
	Order Estimated Delivery ...
Abc	Order Id

Tables

	Order Approved At
	Order Delivered Carrier D...
	Order Delivered Customer...
	Order Estimated Delivery ...
Abc	Order Id
	Order Purchase Timestamp
Abc	Status

d) Perform joins (inner, left, right) on multiple tables

Orders.csv

Orders.csv is made of 1 table. ⓘ

Orders.csv

Orders.csv is made of 2 tables. ⓘ



Orders.csv is made of 2 tables. ⓘ



Join			
Inner	Left	Right	Full Outer
Data Source		Order Payments.csv	
Order Id	=	order id (Order ...	
<i>Add new join clause</i>			

e) Apply data blending between two sources

Search

Filters

Tables

Order Payments.csv

order id (Order Payme...

Payment Type

Orders.csv

Order Approved At

Order Delivered Carrier...

Order Delivered Cust...

Order Estimated Delive...

Order Id

Order Purchase Timest...

Status

Measure Names

Order Payments.csv

Payment Installments

Payment Sequential

Payment Value

Orders.csv (Count)

Measure Values

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Sheet 2

Drop field here

 Orders.csv

Connection

☒ Live ☐ Extract

▼ **Order Payments.csv**

Abc order id (Order Payme...

Abc Payment Type

▼ **Orders.csv**

Abc Customer Id 

 Order Approved At

 Order Delivered Carrier...

 Order Delivered Custa...

 Order Estimated Delive...

Abc Order Id

 Order Purchase Timest...

Abc Status

Blend Relationships

Blend relationships determine how data from secondary data sources are joined with primary data sources.

Primary data source:

Orders

Secondary data source:

☒ Automatic ☐ Custom

Clipboard_20260731T131653

Clipboard_20260731T131655

Customers

Customer Id

Customer Id

Add...

Edit...

Remove

OK

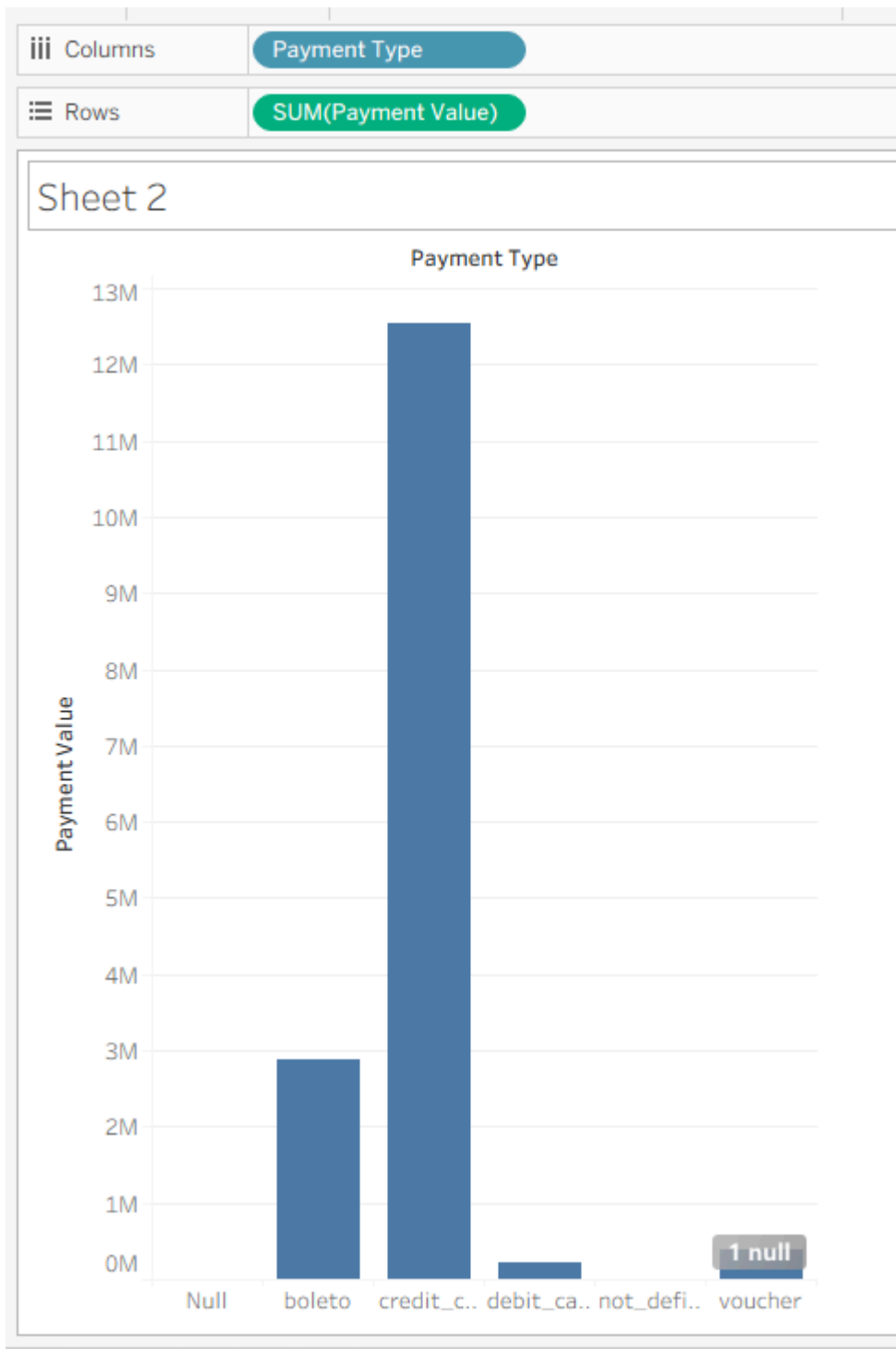
Cancel

Columns	Customer State
Rows	Status

Sheet 2

Status	AC	AL	AM	AP	BA	CE	DF	ES
approved								
canceled		Abc			Abc	Abc	Abc	Abc
created							Abc	
delivered	Abc	Abc	Abc	Abc	Abc	Abc	Abc	Abc
invoiced		Abc			Abc	Abc	Abc	Abc
processing		Abc		Abc	Abc	Abc	Abc	Abc
shipped	Abc	Abc	Abc		Abc	Abc	Abc	Abc
unavailable		Abc	Abc		Abc	Abc	Abc	Abc

f) Implement different types of filters (dimension, measure, date)



drag payment type to filters pane

Filter [Payment Type]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

- ☐ Null
- ☒ boleto
- ☒ credit_card
- ☐ debit_card
- ☐ not_defined
- ☐ voucher

All None ☐ Exclude

Summary

Field: [Payment Type]
Selection: Selected 2 of 6 values
Wildcard: All
Condition: None
Limit: None

Reset OK Cancel Apply

Filter [Payment Value]

Range of values At least At most Special

At least

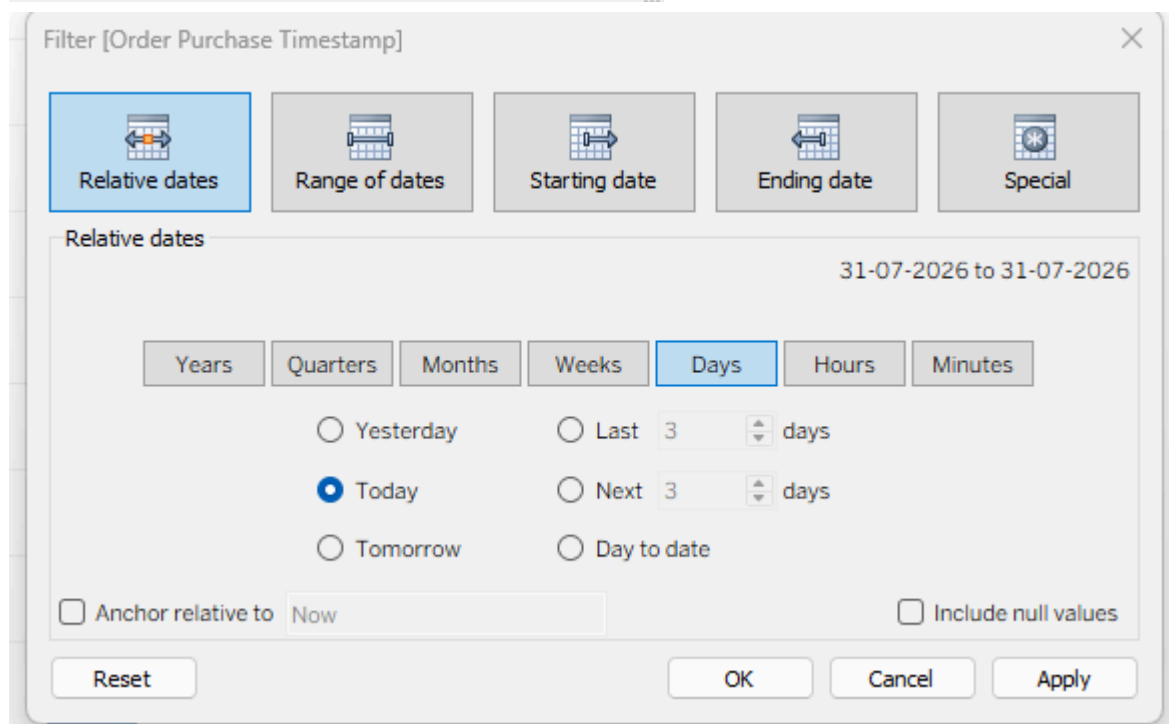
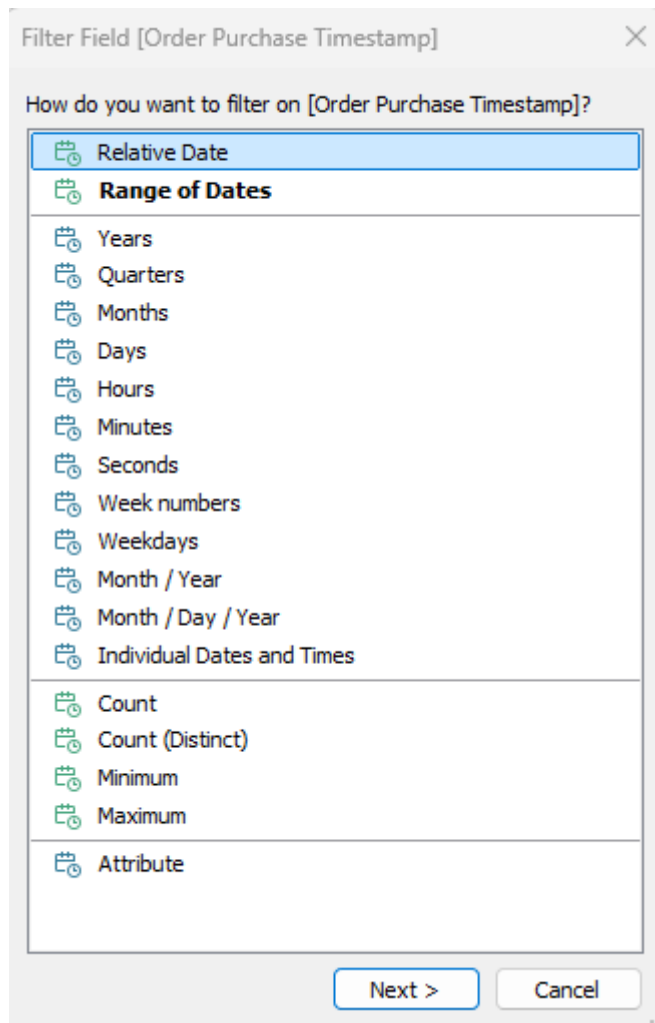
50 13,664.08

0.01 13664.08

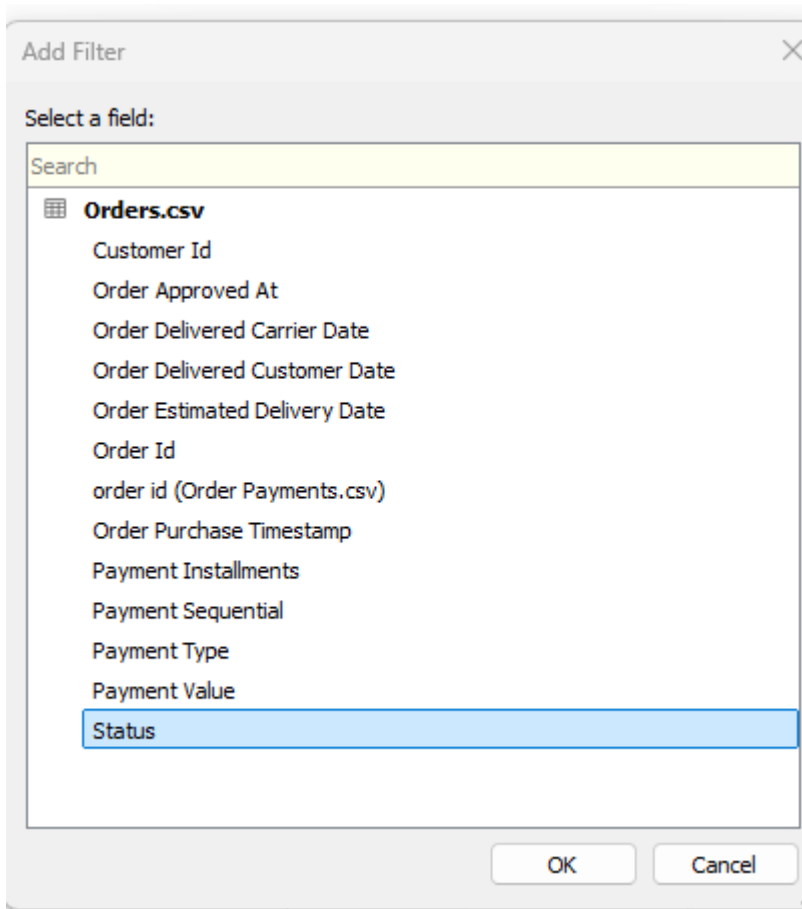
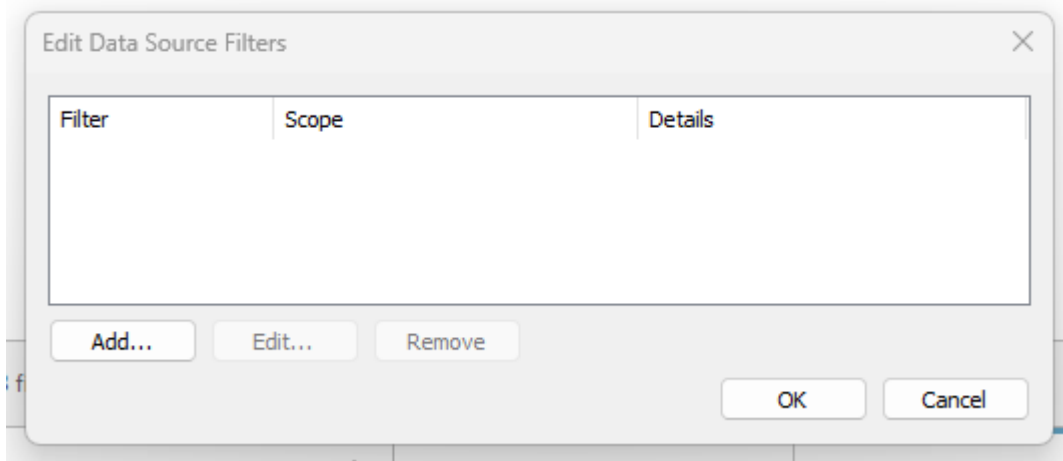
Show: Only Relevant Values ☐ Include Null Values

Reset OK Cancel Apply

drag orders purchase timestamp to filters pane



g) Optimize performance using extracts and filtering
Data source-> on right side click on add



Filter [Status]

General

Wildcard

Condition

Top

☒ Select from list

☐ Custom value list

☐ Use all

Enter search text

☐ approved

☐ canceled

☐ created

☒ delivered

☐ invoiced

☐ processing

☐ shipped

☐ unavailable

All

None

☐ Exclude

Summary

Field: [Status]

Selection: Selected 1 of 8 values

Wildcard: All

Condition: None

Limit: None

Reset

OK

Cancel

Edit Data Source Filters

Filter	Scope	Details
Status	Orders.csv and related tables	keeps delivered

Add...

Edit...

Remove

OK

Cancel

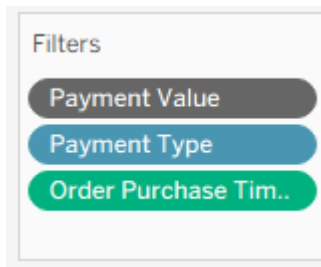
Filters

Payment Type

Payment Value

Order Purchase Tim..

right click on payment value->add the context



Part C – Curiosity Question

Curiosity Question

1. Draw a diagram showing how Tableau connects to Excel, CSV, and Database data sources. Explain the process in two lines.

2. **A company wants to see real-time sales data. Which connection (Live or Extract) would you recommend and why?**
3. **Draw an example of an Inner Join between two tables and explain when it is used.**
4. **A sales manager wants to analyze only the sales made in the year 2025. Which type of filter would you apply and why?**
5. **Draw a simple diagram of data blending using two data sources and explain one practical application.**